

Architecture of Wireless Sensor Networks

Outline:

1. **Hardware Unit of a Sensor Node**
2. **Sensor Network Architecture**
3. **Energy Consumption of Sensor Node**
4. **Optimization Goal and Figure of Merit**

Hardware Unit of a Sensor Node

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A wireless sensor device is generally composed of four basic components:

- Sensing Unit
- Processing Unit
- Transceiver Unit
- Power Unit

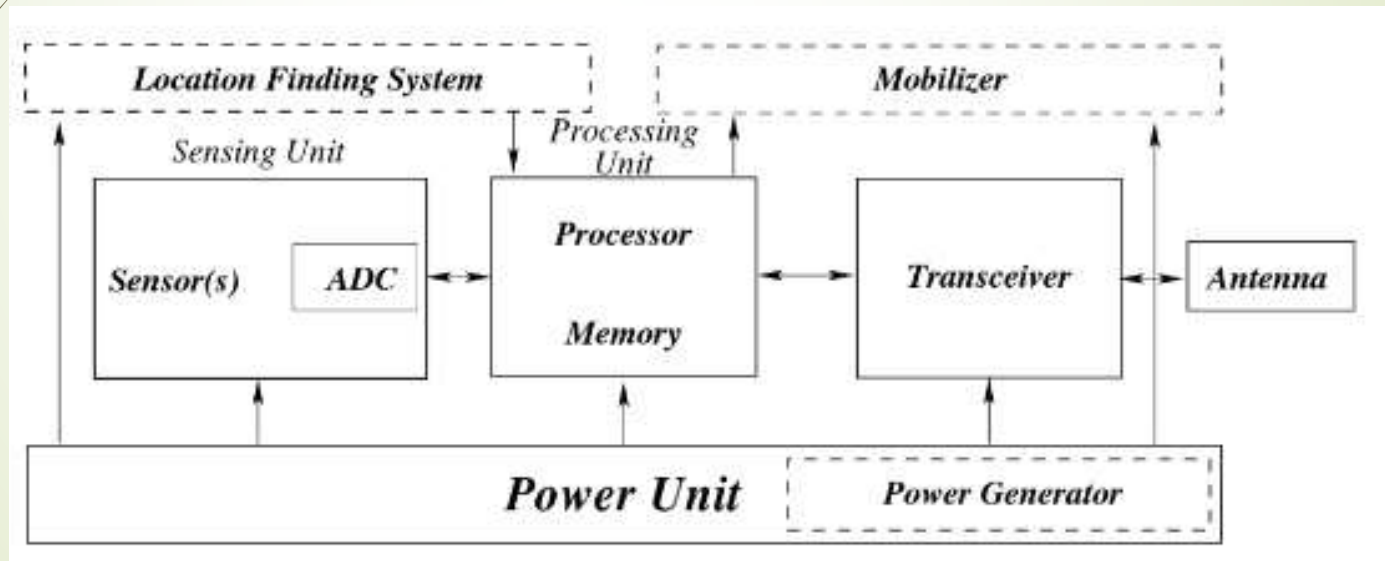


Fig.2.1 General hardware architecture of a sensor node

Hardware Unit of a Sensor Node

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Sensing Unit:

- Main component of a wireless sensor node that distinguishes it from any other embedded system with communication capabilities.
- Sensor unit provides information from the physical world. e.g., Temperature, Humidity, etc.
- The analog signal produced by the sensor are converted to digital signal by the ADC and fed into the processing unit.

Processing Unit:

- The processing unit is the main controller of the wireless sensor node, through which every other component is managed.
- May consist of an on-board memory or may be associated with a small storage unit integrated into the embedded system.
- Enables the sensor node to perform sensing operations, run associated algorithm and collaborate with the other nodes through wireless communication.

Hardware Unit of a Sensor Node

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Transceiver Unit:

- Perform communication between two wireless sensor node.
- Converts bits into RF wave and recover them at the other end.
- Essentially the WSN is connected to the network through this unit.

Power Unit:

- Each component in the wireless sensor node is powered through the power unit and the limited capacity of this unit requires energy efficient operation for the task performed by each component.

Location Finding System:

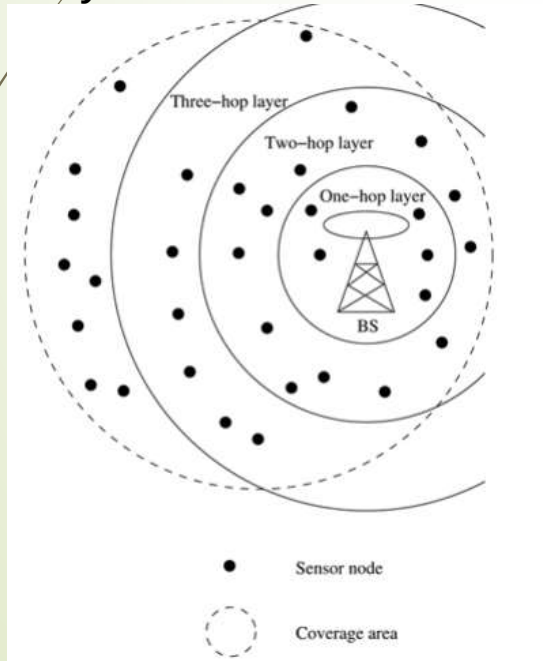
- Physical location of the node is essential.
- System may consist of GPS module or may be a software module which provides the location of the node.

Wireless Sensor Networks Architecture

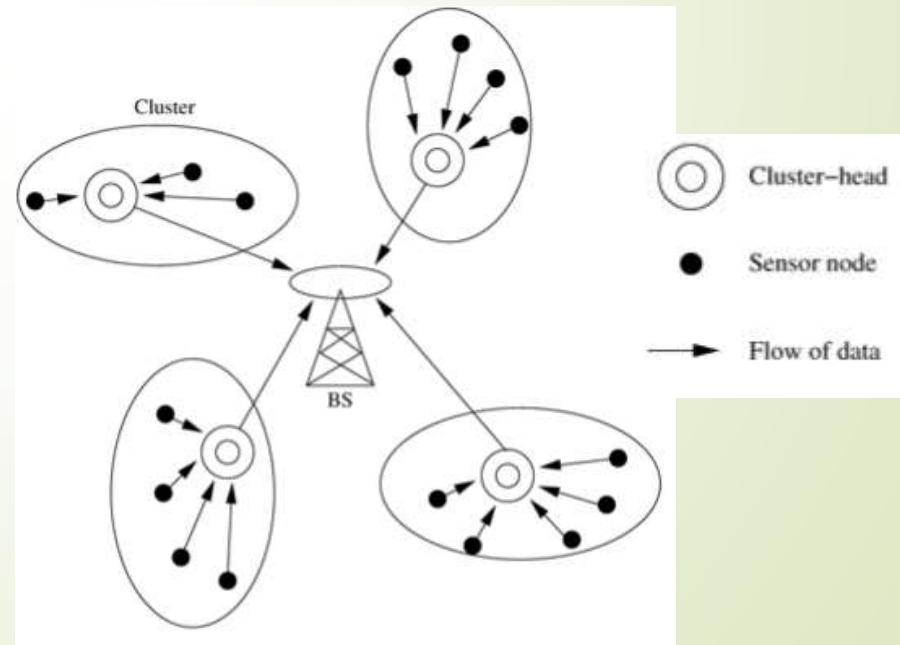
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- The design of sensor networks is influenced by factors such as scalability, fault tolerance, and power consumption.
- The two basic kinds of sensor network architecture are
 - Layered Architecture
 - Clustered Architecture

Layered Architecture



Clustered Architecture



Layered Network Architecture

- This kind of network uses hundreds of sensor nodes as well as a base station.
- Here the arrangement of network nodes can be done into concentric layers.
- Layered architectures have been used with in-building wireless backbones, and in military sensor-based infrastructure, such as the multi-hop infrastructure network architecture.
- The advantage of a layered architecture is that each node is involved only in **short-distance, low-power transmissions** to nodes of the neighboring layers.
- **Unified Network Protocol framework (UNPF)** is a set of protocols for complete implementation of a layered architecture for sensor networks.
- UNPF integrates three operations in its protocol structure: **Network Initialization and Maintenance, MAC, and Routing Protocols.**

Network Initialization and Maintenance Protocol:

- The network initialization protocol organizes the sensor nodes into different layers, using the broadcast capability of the BS.
- The BS can reach all nodes in a one-hop communication over a common control channel.
- The BS broadcasts its identifier (ID) using a known CDMA code on the common control channel.
- All nodes which hear this broadcast then record the BS ID. They send a beacon signal with their own IDs at their low default power levels. Those nodes which the BS can hear form layer one since they are at a single-hop distance from the BS.
- The BS now broadcasts a control packet with all layer one node IDs. All nodes send a beacon signal again.
- The layer one nodes record the IDs which they hear, and these form layer two, since they are one hop away from layer one nodes.
- In the next round of beacons, the layer one nodes inform the BS of the layer two nodes, which is then broadcast to the entire network.
- In this way, the layered structure is built by successive rounds of beacons and BS broadcasts.
- Periodic beaconing updates neighbor information and alters the layer structure if nodes die out or move out of range

MAC Protocol:

- Network initialization is carried out on a common control channel.
- During the data transmission phase, the distributed TDMA receiver oriented channel (DTROC) assignment MAC protocol is used.
- Each node is assigned a reception channel by the BS, and channel reuse is such that collisions are avoided.
- The node schedules transmission slots for all its neighbors and broadcasts the schedule.
- This enables collision-free transmission and saves energy, as nodes can turn off when they are not involved in a send/receive operation.
- The two steps of DTROC are channel allocation (the assignment of reception channels to the nodes) and channel scheduling (the sharing of the reception channel among the neighbors).
- DTROC avoids hidden terminal and exposed terminal problems by suitable channel allocation algorithms.

Routing Protocol:

- Downlink from the BS is by direct broadcast on the control channel.
- The layered architecture enables multi-hop data forwarding from the sensor nodes to the BS.
- The node to which a packet is to be forwarded is selected considering the remaining energy of the nodes.
- This achieves a higher network lifetime.
- Existing ad hoc routing protocols can be simplified for the layered architecture, since only nodes of the next layer need to be maintained in the routing table.

Clustered Architecture

- Clustered architecture is specially useful for sensor networks because of its inherent suitability for data fusion.
- The data gathered by all members of the cluster can be fused at the cluster-head, and only the resulting information needs to be communicated to the BS.
- Sensor networks should be self-organizing, hence the cluster formation and election of cluster-heads must be an autonomous, distributed process.
- This is achieved through network layer protocols such as the **Low-Energy Adaptive Clustering Hierarchy (LEACH)**.

Low-Energy Adaptive Clustering Hierarchy (LEACH)

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- ▶ LEACH is a clustering-based protocol that minimizes energy dissipation in sensor networks.
- ▶ LEACH randomly selects nodes as cluster-heads and performs periodic reelection, so that the high-energy dissipation experienced by the cluster-heads in communicating with the BS is spread across all nodes of the network.
- ▶ Each iteration of selection of cluster-heads is called a round.
- ▶ The operation of LEACH is split into two phases: set-up and steady.
- ▶ During the set-up phase, each sensor node chooses a random number between 0 and 1.
- ▶ If this is lower than the threshold for node n , $T(n)$, the sensor node becomes a cluster-head.
- ▶ This ensures that all sensor nodes eventually spend equal energy. After selection, the cluster-heads advertise their selection to all nodes.
- ▶ All nodes choose their nearest cluster-head when they receive advertisements based on the received signal strength. The cluster-heads then assign a TDMA schedule for their cluster members.

Energy Consumption of Sensor Node:

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- Energy supply for a sensor node is at a premium
- Batteries have small capacity, and recharging by energy scavenging is complicated and volatile.
- Hence, the energy consumption of a sensor node must be tightly controlled.
- The main consumers of energy are the controller, the radio front ends, to some degree the memory, and, depending on the type, the sensors.
- The following components involved in energy consumption are,
 - Operation states with different power consumption
 - Microcontroller energy consumption
 - Memory
 - Radio transceivers
 - Relationship between computation and communication →
 - Power consumption of sensor and actuators

Operation States With different power consumption:

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- One important contribution to reduce power consumption of these components comes from chip-level and lower technologies:
- Designing low-power chips is the best starting point for an energy-efficient sensor node.
- Introducing and using multiple states of operation with reduced energy consumption in return for reduced functionality is the core technique for energy-efficient wireless sensor node

Microcontroller Energy consumption :

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- Basic power consumption in discrete operation states
Embedded controllers commonly implement the concept of multiple operational states as outlined above; it is also fairly easy to control. Some examples probably best explain the idea. The Intel Strong ARM provides three sleep modes:
- In **normal mode**, all parts of the processor are fully powered. Power consumption is up to **400 mW**.
- In **idle mode**, clocks to the CPU are stopped; clocks that pertain to peripherals are active. Any interrupt will cause return to normal mode. Power consumption is up to **100 mW**.
- In **sleep mode**, only the real-time clock remains active. Wakeup occurs after a timer interrupt and takes up to **160 ms**. Power consumption is up to **50 μ W**.

Memory Energy consumption :

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- In fact, the power needed to drive on-chip memory is usually included in the power consumption numbers given for the controllers.
- Hence, the most relevant part is FLASH memory – in fact, the construction and usage of FLASH memory can heavily influence node lifetime.
- The relevant metrics are the read and write times and energy consumption. All this information is readily available from manufacturers' data sheets and do vary depending on several factors.
- Read times and read energy consumption tend to be quite similar between different types of FLASH memory .
- Writing is somewhat more complicated, as it depends on the granularity with which data can be accessed (individual bytes or only complete pages of various sizes).

Radio Transceiver Energy consumption :

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- A radio transceiver has essentially two tasks: transmitting and receiving data between a pair of nodes.
- Similar to microcontrollers, radio transceivers can operate in different modes, the simplest ones are being turned on or turned off.
- To accommodate the necessary low total energy consumption, the transceivers should be turned off most of the time and only be activated when necessary – they work at a low duty cycle.
- But this incurs additional complexity, time and power overhead that has to be taken into account.
- To understand the energy consumption behavior of radio transceivers and their impact on the protocol design, models for the energy consumption per bit for both sending and receiving are required.

Modeling Energy consumption During Transmission :

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- In principle, the energy consumed by a transmitter is due to two sources:
- one part is due to RF signal generation, which mostly depends on chosen modulation and target distance and hence on the transmission power P_{tx} , that is, the power radiated by the antenna.
- A second part is due to electronic components necessary for frequency synthesis, frequency conversion, filters, and so on. These costs are basically constant. One of the most crucial decisions when transmitting a packet is thus the choice of P_{tx} .

Modeling Energy consumption During Reception :

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- Similar to the transmitter, the receiver can be either turned off or turned on.
- While being turned on, it can either actively receive a packet or can be idle, observing the channel and ready to receive. Evidently, the power consumption while it is turned off is negligible.
- Even the difference between idling and actually receiving is very small and can, for most purposes, be assumed to be zero.

Power Consumption of sensors and actuator :

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- Providing any guidelines about the power consumption of the actual sensors and actuators is next to impossible because of the wide diversity of these devices.
- For some of them – for example, passive light or temperature sensors – the power consumption can perhaps be ignored in comparison to other devices on a wireless node report a power consumption of 0.6 to 1 mA for a temperature sensor

Optimization Goals and Figure of Merit:

- The challenging question is how to optimize a network, how to compare these solutions, how to decide which approach better supports a given application, and how to turn relatively imprecise optimization goals into measurable figures of merit.

1. Quality of service:

- WSNs differ from other conventional communication networks mainly in the type of service they offer. These networks essentially only move bits from one place to another.
- Possibly, additional requirements about the offered Quality of Service (QoS) are made, especially in the context of multimedia applications.
- Such QoS can be regarded as a low-level, networking-device-observable attribute - bandwidth, delay, jitter, packet loss rate - or as a high-level, user-observable, so-called subjective attribute like the perceived quality of a voice communication or a video transmission.
- But just like in traditional networks, high-level QoS attributes in WSN highly depend on the application. Some generic possibilities are:

2. Energy Efficiency:

- Energy is a precious resource in WSN that energy efficiency should therefore make an evident optimization goal.
- It is clear that with an arbitrary amount of energy; most of the QoS metrics can be increased.
- Hence, putting the delivered QoS and the energy required to do so into perspective should give a first, reasonable understanding of the term energy efficiency.
- The most commonly considered aspects are:

Energy per correctly received bit: How much energy, counting all sources of energy consumption at all possible intermediate hops, is spent on average to transport one bit of information (payload) from the source to the destination? This is often a useful metric for periodic monitoring applications.

Energy per reported (unique) event : Similarly, what is the average energy spent to report one event? Since the same event is sometimes reported from various sources, it is usual to normalize this metric to only the unique events.

Delay/energy trade-offs : Some applications can increase energy investment for a speedy reporting of such events. Here, the trade-off between delay and energy overhead is interesting.

Network lifetime : The time for which the network is operational or, put another way, the time during which it is able to fulfill its tasks. It is not quite clear, however, when this time ends.

Time to first node death: When does the first node in the network run out of energy or fail and stop operating?

Network half-life: When have 50% of the nodes run out of energy and stopped operating. Any other fixed percentile is applicable as well.

Time to partition : When the first partition of the network in two (or more) disconnected parts occur. This can be as early as the death of the first node or occur very late if the network topology is robust

Time to loss of coverage: A possible figure of merit is thus the time when for the first time any spot in the deployment region is no longer covered by any node's observations

Time to failure of first event notification: This can be due to an event not being noticed because the responsible sensor is dead or because a partition between source and sink has occurred.

3. Scalability:

The ability to maintain performance characteristics irrespective of the size of the network is referred to as scalability.

4. Robustness:

WSN should exhibit an appropriate robustness.

They should not fail just because a limited number of nodes run out of energy, or because their environment changes, these failures have to be compensated by finding other routes.

References

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